

Lec 59

Grundriss

Lehrstuhl

Lec_59



1. Knapsack Problem

Bag

Patience

5 to 10

Input: 1) capacity of the knapsack or bag let m

2) Profits and weights of n objects

• Output: Find Maximum Profit with weight m or less than m

• Ex:

	obj1	obj2	obj3	obj4
profit	200	180	300	150
Weight	12	8	14	5

, and Capacity of bag or knapsack is 31

P/w

	obj1	obj2	obj3	obj4
P/w	$\frac{200}{12}$	$\frac{180}{8}$	$\frac{300}{14}$	$\frac{150}{5}$
	16.67	22.5	21.42	30

$P=0$

300
200
150

$w=31$

$w = 31 - 14 = 17$

$w = 17 - 12 = 5$

$w = 5 - 5 = 0$

→ According to greedy in knapsack each object more P/w calculate profit per weight for the object which has.

$P=0$

$P = 0 + 150$

$w = 31$

$w = 31 - 5 = 26$

$$696.66$$

$$p = 150 + 180 \quad w = 26 - 8$$

$$p = 330 + 300 \quad w = 18 - 14$$

$$p = 630 + 66.66 \quad w = 4 - 4$$

$$w = 0$$

$$12kg \rightarrow \frac{200}{12}$$

$$1kg \rightarrow \frac{200}{12}$$

$$4kg \rightarrow \frac{200}{12}$$

$$\rightarrow \frac{p_i \times w_i}{\sum w_i} = \frac{200}{3} = 66.33$$

$$p \quad ob_1 \quad ob_2 \quad ob_3 \quad \dots \quad ob_j$$

w
 P/w

Conclusion:

Greedy knapsack will give optimal solution by giving priority to profit and weight both

$\Downarrow$
that's why we are calculating
 P/w for each object and
choosing high P/w value
at each iteration

Algo and TC:

	o_1	o_2	o_3	o_4	...	o_n
p	p_1	p_2	p_3	...	...	p_n
w	w_1	w_2	w_3	...	...	w_n
P/w	p_1/w_1	p_2/w_2	p_3/w_3	...	...	p_n/w_n

$m = 6$

Target: To get max^m profit

Greedy Knapsack ()

{ for($i=1$ to n)
calculate P_i/w_i

$\Rightarrow O(n)$

Sort P_i/w_i in decreasing order

$m=6$

pr=0
While($m > 0$ and $w[i] \leq m$)

{
 $m = m - w[i]$
 $pr = pr + p[i]$

$pr = pr + \frac{p[i] \times w[i]}{w[i]}$

$\Rightarrow O(m)$

	o_1	o_2	o_3	o_4
p	6	9	2	13
w	3	2	1	4
P/w	2	4.5	2	3.25

(13) (9) (2)

$m = 6 \rightarrow 0$

$$2 + n \log n \rightarrow m$$

$$\cancel{H}$$
$$O(n \log n)$$

$$SC = O(n)$$

~~→~~
Tosha P/w.

GATE Question:

GATE | GATE CS 2018 | Question 60

Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in rupees)
1	10	60
2	7	28
3	4	20
4	2	24

Question

$$P/w$$

$$60/10 = 6$$

$$28/7 = 4$$

$$20/4 = 5$$

$$24/2 = 12$$

$$P/w$$

$$60/10 = 6$$

$$28/7 = 4$$

$$20/4 = 5$$

$$24/2 = 12$$

$$P/w$$

$$60/10 = 6$$

$$28/7 = 4$$

$$20/4 = 5$$

$$24/2 = 12$$

such that
 you
 Maximize

$$P=0$$

$$w=16$$

$$P=0+24$$

$$w=16-2$$

$$P=24$$

$$w=14$$

$$P=24+60$$

$$w=14-10$$

$$P=84$$

$$w=4-4$$

$$P=84+20$$

$$P=104$$

$$w=0$$



